Vectors and Vector-Valued Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the region R in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving R about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi \int_a^b (5 - x)\sqrt{x}\, dx$. Find the value of k.

The region R is bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$. We are revolving this region about the vertical line $x = 5$. We will use the method of cylindrical shells. The radius of a cylindrical shell at a given x is the distance from the axis of revolution $x = 5$ to x, which is $5 - x$. The height of the shell is the difference in the y-values of the bounding curves, which is $\sqrt{x} - 0 = \sqrt{x}$. The thickness of the shell is dx. The volume of a single cylindrical shell is.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the vector $\mathbf{v}$ with initial point $P = (1, 2)$ and terminal point $Q = (4, 6)$. What are the components of $\mathbf{v}$?

To find the components of a vector $\mathbf{v}$ with initial point (x_1, y_1) and terminal point (x_2, y_2), we subtract the coordinates of the initial point from the coordinates of the terminal point. That is, $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, the initial point is $P = (1, 2)$ and the terminal point is $Q = (4, 6)$. Therefore, the components of $\mathbf{v}$ are $\langle 4 - 1, 6 - 2 \rangle = \langle 3, 4 \rangle$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the curvature of $\mathbf{r}(t) = \langle 3\cos(t), 3\sin(t), 3t \rangle$ at $t = \pi$.

$\kappa = \frac{||\mathbf{r}' \times \mathbf{r}''||}{||\mathbf{r}'||^3}$. At $t = \pi$: $\mathbf{r}' = \langle 0, -3, 3 \rangle$, $\mathbf{r}'' = \langle 3, 0, 0 \rangle$, giving $\kappa = \frac{1}{6}$.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?

The slope of the tangent line to the graph of $f(x)$ at any point is given by its derivative, $f'(x)$. 1. Find the derivative of $f(x) = x^2 - 2x - 1$ using the power rule: $f'(x) = 2x - 2$ 2. Evaluate the derivative at the x -coordinate of point $P(3, 2)$, which is $x = 3$: $f'(3) = 2(3) - 2 = 6 - 2 = 4$ Thus, the slope of the tangent line at P is 4. Misconception Analysis: - 2 results from mistakenly reading the y-coordinate of point P as the slope, since $f(3) = 2$. - 3 results from taking the x-c.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of c.

The equation of the tangent line to $f(x)$ at $x = 1$ is given as $y = 2x + c$. We are given that $f(1) = 3$. This means the point $(1, 3)$ lies on both the function $f(x)$ and its tangent line at $x = 1$. Since the point $(1, 3)$ lies on the tangent line $y = 2x + c$, we can substitute the coordinates into the equation: $3 = 2(1) + c$ $3 = 2 + c$ $c = 3 - 2$ $c = 1$ The derivative of the function at $x = 1$ is the slope of the tangent line, which is 2. This information is consistent with the problem but not direct.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The region bounded by the curve $y = x^2 - 4$ and the x-axis is rotated about the x-axis. If the volume of the resulting solid is $V = \pi\int_a^b (x^2 - 4)^2 dx$, find the value of $b - a$.

The region is bounded by the curve $y = x^2 - 4$ and the x-axis. To find the limits of integration, we need to find the x-intercepts of the curve, where $y = 0$. Setting $x^2 - 4 = 0$, we get $x^2 = 4$, which gives $x = \pm 2$. Therefore, the bounds of integration are $a = -2$ and $b = 2$. The volume of the solid of revolution is given by $V = \pi\int_a^b [f(x)]^2 dx$. In this case, $f(x) = x^2 - 4$, and the limits are $a = -2$ and $b = 2$. The question asks for the value of $b - a$. $b - a = 2 - (-2) = 2 + 2 = 4$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^2 - 6x + 5$. The function has a local minimum at which value of x?

To find the local minimum of the function $f(x) = x^2 - 6x + 5$, we first find the derivative of the function with respect to x: $f'(x) = 2x - 6$. Next, we set the derivative equal to zero to find the critical points: $2x - 6 = 0$. Solving for x, we get $2x = 6$, which means $x = 3$. To determine if this critical point is a local minimum, we can use the second derivative test. The second derivative of $f(x)$ is $f''(x) = 2$. Since $f''(3) = 2 > 0$, the function has a local minimum at $x = 3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^2 - 5x + 6$. What is the derivative of $f(x)$ with respect to x?

To find the derivative of the function $f(x) = x^2 - 5x + 6$, we apply the power rule for differentiation, which states that the derivative of ax^n is anx^{n-1}. The derivative of a constant is 0. Applying the power rule to each term: 1. The derivative of x^2 is $2x^{2-1} = 2x$. 2. The derivative of $-5x$ is $-5 \cdot 1 \cdot x^{1-1} = -5x^0 = -5$. 3. The derivative of the constant 6 is 0. Combining these, the derivative of $f(x)$ is $f'(x) = 2x - 5 + 0 = 2x - 5$.

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9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The function has a local minimum. What are the coordinates of the local minimum?

The function is a parabola opening upwards, so its local minimum is also its vertex. The x-coordinate of the vertex of a parabola given by $f(x) = ax^2 + bx + c$ is $x = -b/(2a)$. For $f(x) = x^2 - 4x + 3$, we have $a = 1$ and $b = -4$. Therefore, the x-coordinate of the vertex is $x = -(-4)/(2 * 1) = 4/2 = 2$. To find the y-coordinate, we substitute $x = 2$ into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Thus, the coordinates of the local minimum are (2, -1).

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the graph of the function $f(x)$ shown below. What is the limit of $f(x)$ as x approaches 1?

The limit of a function as x approaches a certain value is the value that $f(x)$ gets arbitrarily close to as x gets arbitrarily close to that value. Observing the graph, as x approaches 1 from both the left and the right, the value of $f(x)$ approaches 1. Therefore, $\lim_{x \to 1} f(x) = 1$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $z = f(x, y) = x^2 + 2y^2$. What is the value of the partial derivative $\frac{\partial f}{\partial x}$ at the point $(1, 2)$?

To find the partial derivative of $f(x, y) = x^2 + 2y^2$ with respect to x, we treat y as a constant and differentiate with respect to x. The derivative of x^2 with respect to x is $2x$, and the derivative of $2y^2$ (a constant with respect to x) is 0. Therefore, $\frac{\partial f}{\partial x} = 2x$. Now, we evaluate this partial derivative at the point $(1, 2)$. Substituting $x = 1$ into the expression for $\frac{\partial f}{\partial x}$: $\frac{\partial f}{\partial x}$ at $(1, 2) = 2 \times 1 = 2$. Th.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the region bounded by the curve $y = x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. What is the area of this region?

The area of a region bounded by a curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$ is given by the definite integral $\int_a^b f(x)\,dx$. In this case, $f(x) = x^2$, $a = 1$, and $b = 3$. The area A is given by: $A = \int_1^3 x^2\,dx$ To evaluate this integral, we find the antiderivative of x^2, which is $\frac{x^3}{3}$. Now, we apply the Fundamental Theorem of Calculus.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $z = f(x, y) = x^2 - 3xy + 2y^2$. What is the partial derivative of f with respect to y, denoted as $\frac{\partial f}{\partial y}$?

To find the partial derivative of $f(x, y) = x^2 - 3xy + 2y^2$ with respect to y, we treat x as a constant. \\ \\ The derivative of x^2 with respect to y is 0, as x is considered constant. \\ \\ The derivative of $-3xy$ with respect to y is $-3x \cdot 1 = -3x$. \\ \\ The derivative of $2y^2$ with respect to y is $2 \cdot 2y = 4y$. \\ \\ Therefore, the partial derivative $\frac{\partial f}{\partial y}$ is $0 - 3x + 4y = -3x + 4y$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle's position is described by the parametric equations $x(t) = 3t + 1$ and $y(t) = 2t - 2$. What are the coordinates of the particle when $t = 3$?

To find the coordinates of the particle at a specific time t, substitute the value of t into the parametric equations for x and y. Given $x(t) = 3t + 1$ and $y(t) = 2t - 2$. We need to find the coordinates when $t = 3$. For the x-coordinate: $x(3) = 3(3) + 1 = 9 + 1 = 10$ For the y-coordinate: $y(3) = 2(3) - 2 = 6 - 2 = 4$ Therefore, the coordinates of the particle when $t = 3$ are (10, 4).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Area Between Curves

Reduction formula: $\int \sin^n x\,dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} x\,dx$. Then $\int_0^{\pi/2} \sin^6 x\,dx =$

Wallis: $\frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{5\pi \cdot 3}{6 \cdot 4 \cdot 2 \cdot 2} = \frac{15\pi}{96} = \frac{5\pi}{32}$. Actually Wallis: $\int_0^{\pi/2} \sin^{2n} x\,dx = \frac{(2n-1)!!}{(2n)!!}\frac{\pi}{2}$. For $n = 3$: $\frac{5 \cdot 3 \cdot 1}{6 \cdot 4 \cdot 2}\frac{\pi}{2} = \frac{15}{48}\frac{\pi}{2} = \frac{15\pi}{96} = \frac{5\pi}{32}$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.02 Volumes of Revolution

Level 3 · Advanced For the position vector $\mathbf{r}(t) = \langle t, t^2, \frac{2}{3}t^3 \rangle$, find the tangential component of acceleration a_T and the normal component a_N at $t = 1$.

Compute derivatives: $\mathbf{r}'(t) = \langle 1, 2t, 2t^2 \rangle$ $\mathbf{r}''(t) = \langle 0, 2, 4t \rangle$ At $t = 1$: $\mathbf{r}'(1) = \langle 1, 2, 2 \rangle \Rightarrow |\mathbf{r}'(1)| = \sqrt{1 + 4 + 4} = 3$ $\mathbf{r}''(1) = \langle 0, 2, 4 \rangle$ Tangential acceleration a_T: $a_T = \frac{\mathbf{r}' \cdot \mathbf{r}''}{|\mathbf{r}'|} = \frac{(1)(0) + (2)(2) + (2)(4)}{3} = \frac{0 + 4 + 8}{3} = \frac{12}{3} = 4$ Normal acceleration a_N: Compute cross product at $t = 1$.

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17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.03 Arc Length and Improper Integrals**

Level 2 · Medium Find the area of the triangle with vertices $A(1, 1, 0)$, $B(3, 0, 2)$, and $C(0, 2, 4)$.

Find vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = \langle 2, -1, 2 \rangle$ $\vec{AC} = \langle -1, 1, 4 \rangle$ Compute the cross product: $\vec{AB} \times \vec{AC} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 2, -1, 2; -1, 1, 4| = \mathbf{i}(-4-2) - \mathbf{j}(8-(-2)) + \mathbf{k}(2-1) = \langle -6, -10, 1 \rangle$ Magnitude of cross product: $|\vec{AB} \times \vec{AC}| = \sqrt{(-6)^2 + (-10)^2 + 1^2} = \sqrt{36 + 100 + 1} = \sqrt{137}$ The triangle area is half the parallelogram area.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Separable Differential Equations**

Level 1 · Foundations Determine whether the vectors $\mathbf{u} = \langle 2, -3, 4 \rangle$ and $\mathbf{v} = \langle 6, 8, 3 \rangle$ are orthogonal, parallel, or neither.

Compute the dot product: $\mathbf{u} \cdot \mathbf{v} = (2)(6) + (-3)(8) + (4)(3) = 12 - 24 + 12 = 0$ Since the dot product is exactly 0 and neither vector is the zero vector, **u** and **v** are strictly orthogonal (perpendicular).

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Precalculus Review for Calculus**

[Vectors & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves**

Level 2 · MediumOn Section II Part A (Calculator Active), a student is asked for the acceleration vector at $t = 3$ for a particle whose velocity is $\vec{v}(t) = \langle \ln(t^2 + 1), 3\cos(t) \rangle$. What calculator command must be used to find this vector, and what is its 3-decimal numerical value?

Acceleration is the derivative of velocity: $\vec{a}(3) = \langle x''(3), y''(3) \rangle$. Use the calculator's numerical derivative function nDeriv: • $x''(3) = \frac{d}{dt}[\ln(t^2 + 1)]\Big|_{t=3} = \frac{2(3)}{3^2 + 1} = \frac{6}{10} = 0.6$. • $y''(3) = \frac{d}{dt}[3\cos(t)]\Big|_{t=3} = -3\sin(3) \approx -0.423$. Thus: $\mathbf{\vec{a}(3) = \langle 0.600, -0.423 \rangle}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the cross product $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = \langle 1, 0, 2 \rangle$ and $\mathbf{v} = \langle 0, 3, -1 \rangle$.

a. $\langle -6, 1, 3 \rangle$ ✓

b. $\langle 6, -1, -3 \rangle$

c. $\langle -6, -1, 3 \rangle$

d. $\langle 6, 1, 3 \rangle$

$\mathbf{u} \times \mathbf{v} = \langle 0(-1) - 2(3), 2(0) - 1(-1), 1(3) - 0(0) \rangle = \langle -6, 1, 3 \rangle$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the area of the parallelogram determined by $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 3, 4 \rangle$.

a. $\sqrt{89}$ ✓

b. 9

c. $\sqrt{73}$

d. 10

$\mathbf{u} \times \mathbf{v} = \langle 8, -4, 3 \rangle$. Magnitude $||\mathbf{u} \times \mathbf{v}|| = \sqrt{64 + 16 + 9} = \sqrt{89}$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the unit tangent vector $\mathbf{T}(t)$ to $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$.

a. $\langle -\frac{3}{5}\sin t, \frac{3}{5}\cos t, \frac{4}{5} \rangle$ ✓

b. $\langle -3\sin t, 3\cos t, 4 \rangle$

c. $\langle -\frac{3}{4}\sin t, \frac{3}{4}\cos t, 1 \rangle$

d. $\frac{1}{5}\langle 3\cos t, 3\sin t, 4t \rangle$

$\mathbf{r}'(t) = \langle -3\sin t, 3\cos t, 4 \rangle$. $||\mathbf{r}'(t)|| = \sqrt{9\sin^2 t + 9\cos^2 t + 16} = \sqrt{25} = 5$. Thus $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{5}$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the arc length of the circular helix $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$ for $0 \le t \le 2\pi$.

a. 10π ✓

b. 5π

c. 20π

d. 6π

Speed $||\mathbf{r}'(t)|| = \sqrt{(-3\sin t)^2 + (3\cos t)^2 + 4^2} = \sqrt{9 + 16} = 5$. Arc length $L = \int_0^{2\pi} 5\, dt = 10\pi$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the formula for curvature $\kappa(t)$ of a space curve in terms of $\mathbf{r}'$ and $\mathbf{r}''$.

a. $\kappa(t) = \frac{||\mathbf{r}'(t) \times \mathbf{r}''(t)||}{||\mathbf{r}'(t)||^3}$ ✓

b. $\kappa(t) = \frac{||\mathbf{r}''(t)||}{||\mathbf{r}'(t)||^2}$

c. $\kappa(t) = \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{||\mathbf{r}'(t)||^3}$

d. $\kappa(t) = ||\mathbf{r}''(t)||$

The standard vector formula for the curvature of a twice-differentiable space curve is $\kappa = \frac{||\mathbf{r}' \times \mathbf{r}''||}{||\mathbf{r}'||^3}$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the curvature of a circle of radius R.

a. $\kappa = 1/R$ ✓

b. $\kappa = R$

c. $\kappa = 2\pi R$

d. $\kappa = 1/R^2$

A circle of radius R has constant curvature equal to the reciprocal of its radius: $\kappa = 1/R$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If position is $\mathbf{r}(t) = \langle t^2, 4t, -16t^2 + 64t \rangle$, find the velocity and acceleration vectors at $t = 1$.

a. $\mathbf{v}(1) = \langle 2, 4, 32 \rangle, \mathbf{a}(1) = \langle 2, 0, -32 \rangle$ ✓

b. $\mathbf{v}(1) = \langle 1, 4, 48 \rangle, \mathbf{a}(1) = \langle 2, 0, -16 \rangle$

c. $\mathbf{v}(1) = \langle 2, 4, 0 \rangle, \mathbf{a}(1) = \langle 2, 0, -32 \rangle$

d. $\mathbf{v}(1) = \langle 2, 4, -32 \rangle$

$\mathbf{v}(t) = \mathbf{r}'(t) = \langle 2t, 4, -32t + 64 \rangle$. At $t = 1$: $\mathbf{v}(1) = \langle 2, 4, 32 \rangle$. $\mathbf{a}(t) = \langle 2, 0, -32 \rangle$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the speed at $t = 0$ for a particle with velocity $\mathbf{v}(t) = \langle 3\cos t, 4, 3\sin t \rangle$.

a. 5 ✓

b. 7

c. 4

d. $\sqrt{7}$

Speed is the magnitude of velocity: $||\mathbf{v}(0)|| = \sqrt{3^2 + 4^2 + 0^2} = \sqrt{9 + 16} = 5$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Decompose acceleration into tangential and normal components: $\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}$. What is a_T?

a. $a_T = \frac{\mathbf{v}\cdot\mathbf{a}}{||\mathbf{v}||}$ ✓

b. $a_T = \frac{||\mathbf{v}\times\mathbf{a}||}{||\mathbf{v}||}$

c. $a_T = ||\mathbf{a}||$

d. $a_T = \frac{\mathbf{a}}{||\mathbf{v}||}$

The tangential component of acceleration represents the rate of change of speed: $a_T = \frac{d}{dt}||\mathbf{v}|| = \frac{\mathbf{v}\cdot\mathbf{a}}{||\mathbf{v}||}$.

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the arc length of $\mathbf{r}(t) = \langle 2t, t^2, \ln(t)\rangle$ for $1 \le t \le 3$.

a. $2\ln(3) + 16$

b. $\ln(3) + 8$ ✓

c. $\frac{\ln(3)}{2} + 4$

d. $\ln(3) + 9$

$L = \int_a^b ||\mathbf{r}'(t)||\,dt$; here $||\mathbf{r}'(t)||$ simplifies to a function with an elementary antiderivative, and the integral equals $\ln(3) + 8$.

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Arc length of $\mathbf{r}(t) = (e^t\cos t, e^t\sin t, e^t)$ from $t = 0$ to $t = 1$:

a. $e - 1$

b. $\sqrt{3}(e-1)$ ✓

c. $\sqrt{2}(e-1)$

d. $e\sqrt{3}$

$|r'| = \sqrt{3}e^t \Rightarrow L = \sqrt{3}(e-1)$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For $\mathbf{r}(t) = \langle 2t, 3t^2, t\rangle$, find the normal component of acceleration at $t = 1$.

a. $\frac{6\sqrt{205}}{41}$ ✓

b. $\frac{36\sqrt{41}}{41}$

c. 6

d. 36

$\mathbf{v} = \langle 2, 6, 1\rangle$, $\mathbf{a} = \langle 0, 6, 0\rangle$. $a_N = \frac{||\mathbf{v} \times \mathbf{a}||}{||\mathbf{v}||} = \frac{6\sqrt{205}}{41}$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The magnitude of cross product $|\mathbf{a} \times \mathbf{b}|$ equals the area of the parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$. For $|a| = 3, |b| = 4, \theta = \pi/6$:

a. 24

b. 12

c. 3

d. 6 ✓

Applying Vectors and Vector-Valued Functions principles, $|a||b|\sin(\pi/6) = 12 \cdot 1/2 = 6$. Therefore, the verified correct option is 6.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Cross product $\mathbf{a} \times \mathbf{b}$ is orthogonal to:

a. $\mathbf{a}$ only

b. $\mathbf{b}$ only

c. neither

d. **both $\mathbf{a}$ and $\mathbf{b}$** ✓

$\mathbf{a} \times \mathbf{b} \perp \mathbf{a}$ and $\perp \mathbf{b}$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Flux of $\mathbf{F} = (x, y, 0)$ through lateral surface of cylinder $x^2 + y^2 = 1$, $0 \le z \le 2$, outward:

a. 0

b. 2π

c. 4π ✓

d. π

$abla \cdot \mathbf{F} = 2$; by divergence theorem on cylinder: $2 \cdot \text{Vol} = 2 \cdot 2\pi - \text{caps}$. Easier: direct: $\hat{n} = (\cos\theta, \sin\theta, 0)$, $\mathbf{F} = (\cos\theta, \sin\theta, 0) \Rightarrow \mathbf{F} \cdot \hat{n} = 1$; $\iint_S 1\, dS = 2\pi \cdot 2 = 4\pi$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Parametric surface $\mathbf{r}(u, v) = (u\cos v, u\sin v, u^2)$. At $(u, v) = (1, 0)$:

a. tangent plane is $z = x^2$

b. **tangent plane is $z = 2x - 1$** ✓

c. normal is $(0, 0, 1)$

d. surface is a sphere

At $(u, v) = (1, 0)$, the point is $(1, 0, 1)$, $\mathbf{r}_u = (1, 0, 2)$, and $\mathbf{r}_v = (0, 1, 0)$, giving normal $\mathbf{r}_u \times \mathbf{r}_v = (-2, 0, 1)$ and tangent plane $-2(x - 1) + z - 1 = 0 \Rightarrow z = 2x - 1$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Vector triple product: $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) =$

a. $\mathbf{0}$

b. $(\mathbf{a} \cdot \mathbf{b})\mathbf{c} - (\mathbf{a} \cdot \mathbf{c})\mathbf{b}$

c. $\mathbf{b}(\mathbf{a} \cdot \mathbf{c}) + \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$

d. $(\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$ ✓

Applying Vectors and Vector-Valued Functions principles, BAC-CAB rule. Therefore, the verified correct option is $(\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Separable Differential Equations

Which is an equation of the plane through $(-3,\ 1,\ 4)$, $(-2,\ 2,\ 4)$ and $(1,\ -1,\ 1)$?

a. $x - y + 2z = -4$

b. $x + y + 2z = 6$

c. $x - y + 2z = 4$ ✓

d. $x + y = -2$

A normal vector is $\vec{PQ} \times \vec{PR}$, a multiple of $\langle 1,\ -1, 2\rangle$. Using the point $(-3,\ 1,\ 4)$ gives $x - y + 2z = 4$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Precalculus Review for Calculus

Find the work done by $\mathbf{F} = (x^2, y^2, z^2)$ along line segment from $(0, 0, 0)$ to $(1, 1, 1)$:

a. 1 ✓

b. $\frac{2}{3}$

c. 3

d. $\frac{1}{2}$

Parameterizing the segment by $\mathbf{r}(t) = (t, t, t)$ for $0 \le t \le 1$, the work is $W = \int_0^1 (t^2 + t^2 + t^2)\,dt = 3\left[\frac{t^3}{3}\right]_0^1 = 1$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Area of surface $z = f(x, y)$ over D:

a. $\iint_D f\,dA$

b. $\iint_D \sqrt{1 + f_x^2 + f_y^2}\,dA$ ✓

c. $\iint_D (f_x + f_y)dA$

d. $\iint_D f^2\,dA$

Applying Vectors and Vector-Valued Functions principles, Surface area formula. Therefore, the verified correct option is $\iint_D \sqrt{1 + f_x^2 + f_y^2}\,dA$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Sequences and Series

For $\mathbf{r}(t) = \langle 2t, t^2, t\rangle$, find the tangential component of acceleration at $t = 2$.

a. $\frac{2\sqrt{105}}{21}$

b. $\frac{8\sqrt{21}}{21}$ ✓

c. 2

d. 8

$\mathbf{v} = \langle 2, 4, 1\rangle$, $\mathbf{a} = \langle 0, 2, 0\rangle$. $a_T = \frac{\mathbf{v}\cdot\mathbf{a}}{||\mathbf{v}||} = \frac{8\sqrt{21}}{21}$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 12.01 Convergence Tests

The torsion τ of a space curve measures how fast the curve twists out of its osculating plane. For a helix $\mathbf{r} = (a\cos t, a\sin t, bt)$, as $b \to 0$ (curve flattens to a circle), $\tau \to$

a. $1/a$

b. 0 ✓

c. ∞

d. b

Applying Vectors and Vector-Valued Functions principles, $\tau = b/(a^2 + b^2) \to 0$ as $b \to 0$. Therefore, the verified correct option is 0.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 12.02 Power Series and Taylor Series

Find the area of the triangle with vertices $(1,\ 0,\ 2), (1,\ 1,\ 2), (-3,\ 3,\ -3)$.

a. $\sqrt{41}$

b. $\frac{\sqrt{41}}{3}$

c. $\frac{\sqrt{41}}{2}$ ✓

d. $\frac{3}{2}$

$\vec{PQ} \times \vec{PR} = \langle -5, 0, 4 \rangle$, whose length is $\sqrt{41}$. The area is half of that: $\frac{\sqrt{41}}{2}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

The normal vector to the surface $F(x, y, z) = 0$ at a point is:

a. $(F_x, F_y, F_z) = ablaF$ ✓

b. (F_{xx}, F_{yy}, F_{zz})

c. the Hessian of F

d. $(F_y, F_x, 0)$

Applying Vectors and Vector-Valued Functions principles, $ablaF \perp$ level set. Therefore, the verified correct option is $(F_x, F_y, F_z) = ablaF$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 15.01 Double Integrals

Distance from point $P = (1, 2, 3)$ to plane $2x + y - 2z = 5$:

a. $\frac{7}{3}$ ✓

b. $\frac{2}{3}$

c. 1

d. 3

Using the point-to-plane distance formula, $d = \frac{|2(1) + 1(2) - 2(3) - 5|}{\sqrt{2^2 + 1^2 + (-2)^2}} = \frac{|-7|}{3} = \frac{7}{3}$.

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